Stochastic Mathematics for Finance
Week 3 Resources

Purpose of Week 3

In Week 2, we studied Gaussian behaviour, the Central Limit Theorem, and Monte Carlo methods. We observed how sums of many independent random variables converge toward Gaussian distributions, and how random sampling can approximate mathematical quantities numerically.

This week, we take the next fundamental step: moving from discrete random walks and static probability to **continuous-time stochastic processes**. This is the transition from probability theory to stochastic calculus.

The central object of study is **Standard Brownian Motion**, also called the Wiener process. Brownian motion is the continuous-time limit of a symmetric random walk, and it forms the mathematical foundation of modern stochastic calculus and quantitative finance. Every continuous-time financial model in use today — from Black–Scholes to stochastic volatility to interest rate models — is built on top of Brownian motion.

We will study the key properties of Brownian motion, simulate its paths computationally, and develop intuition for **quadratic variation**, which is one of the most important concepts distinguishing stochastic calculus from ordinary calculus. The fact that Brownian motion has non-zero quadratic variation is the reason ordinary calculus breaks down and a new calculus must be invented.

We will also introduce **Itô's Lemma**, the fundamental chain rule of stochastic calculus. Itô's Lemma tells us how smooth functions of Brownian motion evolve over time, and it underlies the derivation of virtually every major result in mathematical finance, including the Black–Scholes equation.

Finally, we will introduce the **Euler–Maruyama method**, a numerical scheme for simulating stochastic differential equations (SDEs), and apply it to basic models including Geometric Brownian Motion and the Ornstein–Uhlenbeck process.

By the end of this week, you should be comfortable with:

- the definition and properties of Standard Brownian Motion W_t;
- independent and stationary increments;
- the concept of a filtration and what it means for a process to be adapted;
- the Gaussian structure of Brownian paths and their covariance;
- total variation versus quadratic variation;

- quadratic variation and the identity $dW^2 = dt$;
- the contrast between quadratic variation of Brownian motion and smooth functions;
- the non-differentiability of Brownian paths and why it matters;
- what a martingale is and why Brownian motion is one;
- the Itô stochastic integral and how it differs from the Riemann integral;
- Itô's Lemma and its intuitive derivation from Taylor expansion;
- applying Itô's Lemma to basic functions;
- stochastic differential equations: drift, diffusion, and the SDE framework;
- the Euler–Maruyama numerical method for SDEs;
- simulating Brownian motion paths in Python;
- empirically verifying the non-zero quadratic variation of simulated paths.

Contents

1. From Random Walks to Brownian Motion

1.1. Recap: The Discrete Random Walk

In Week 1 we studied random walks on the integers. Recall that the simple symmetric random walk is defined by

$$S_n = X_1 + X_2 + \cdots + X_n,$$

where $X_1, X_2, \ldots$ are iid random variables satisfying

$$P(X_i = +1) = P(X_i = -1) = \frac{1}{2}.$$

From Week 2, we know that

$$E[S_n] = 0, \qquad \mathrm{Var}(S_n) = n,$$

and by the Central Limit Theorem,

$$\frac{S_n}{\sqrt{n}} \Rightarrow \mathcal{N}(0,1).$$

The random walk is a useful model but it lives in discrete time: it only takes values at integer times $n = 0, 1, 2, \ldots$. Financial markets, physical phenomena, and most real-world processes evolve in continuous time. We need a process that:

- lives at every time $t \in [0, \infty)$, not just at integers;
- has the same Gaussian and independence structure as the random walk;
- has continuous sample paths.

Brownian motion is exactly this object.

1.2. Constructing the Continuous Limit

Fix a time horizon $T > 0$ and divide it into n equal steps of size

$$\Delta t = \frac{T}{n}.$$

At each step, the walk moves by $\pm\sqrt{\Delta t}$. The scaling by $\sqrt{\Delta t}$ rather than 1 is crucial: it ensures the variance of the process over any fixed time interval $[0, T]$ remains T regardless of how finely we divide the interval.

Define the rescaled process at continuous time $t \in [0, T]$ by

$$W_t^{(n)} = \sqrt{\Delta t} \sum_{k=1}^{\lfloor t/\Delta t \rfloor} X_k.$$

As $n \to \infty$ (equivalently $\Delta t \to 0$), the process $W_t^{(n)}$ converges in distribution to a continuous-time limit. That limit is Standard Brownian Motion.

To see why the variance works out correctly, note that for any interval $[s, t]$, the number of steps taken is approximately $(t - s)/\Delta t$, so the increment has variance

$$\frac{t-s}{\Delta t} \cdot (\sqrt{\Delta t})^2 = t - s.$$

This is independent of Δt, confirming that the variance of the limiting process over $[s, t]$ equals $t - s$, which is the key property of Brownian motion.

1.3. Why Brownian Motion Matters

Brownian motion is the foundation of stochastic calculus and mathematical finance because:

- it is the simplest continuous-time stochastic process with independent increments;
- it has well-understood Gaussian structure that can be fully characterized;
- it drives the randomness in virtually all continuous-time financial models;
- it is the building block for more complex processes such as Geometric Brownian Motion, Ornstein–Uhlenbeck processes, and stochastic volatility models;
- the mathematical theory of stochastic integration is built around it.

Historically, Brownian motion was first observed physically by the botanist Robert Brown in 1827, who noticed the erratic movement of pollen particles suspended in water. It was given a rigorous mathematical treatment by Norbert Wiener in the 1920s, which is why it is also called the Wiener process. Louis Bachelier used it to model stock prices as early as 1900, predating Einstein's famous 1905 paper on the subject.

2. Filtrations and Information

Before giving the formal definition of Brownian motion, we need to introduce a concept that is entirely new to this course: the **filtration**. This concept is central to all of continuous-time probability and finance, and it did not appear in Weeks 1 or 2.

2.1. What is a Filtration?

In probability theory, a **filtration** $\{\mathcal{F}_t\}_{t \geq 0}$ is an increasing family of σ-algebras representing the information available up to time t.

Concretely, $\mathcal{F}_t$ is the collection of all events whose occurrence or non-occurrence is known by time t. As time passes, information accumulates and never disappears:

$$s \leq t \implies \mathcal{F}_s \subseteq \mathcal{F}_t.$$

Think of $\mathcal{F}_t$ as the answer to the question: *what do we know at time t?*

2.2. The Natural Filtration of Brownian Motion

For Brownian motion, the **natural filtration** is defined by

$$\mathcal{F}_t = \sigma(W_s : 0 \le s \le t),$$

which is the smallest σ-algebra that makes all the random variables $\{W_s : s \le t\}$ measurable. In plain language, $\mathcal{F}_t$ contains all information that can be deduced from observing the path of W up to time t.

2.3. Adapted Processes

A stochastic process $\{X_t\}_{t\ge 0}$ is called **adapted** to the filtration $\{\mathcal{F}_t\}$ if X_t is $\mathcal{F}_t$-measurable for each $t \ge 0$.

Intuitively, an adapted process is one whose value at time t depends only on information available up to time t — it cannot see into the future.

This is a fundamental requirement in finance. A trading strategy must be adapted: the number of shares held at time t can only depend on prices observed up to time t, not on future prices. This is the mathematical formalization of the no-clairvoyance principle.

2.4. Why Filtrations Matter

The language of filtrations allows us to make precise statements about:

- **Independence.** The increment $W_t - W_s$ is independent of $\mathcal{F}_s$, meaning the future movement of Brownian motion is independent of its past.
- **Conditional expectations.** We can write $E[X \mid \mathcal{F}_t]$ to mean the expected value of X given everything known at time t.
- **Martingales.** A process is a martingale if its expected future value, given current information, equals its current value.

3. Standard Brownian Motion

3.1. Formal Definition

A stochastic process $\{W_t\}_{t\ge 0}$ defined on a probability space $(\Omega, \mathcal{F}, P)$ equipped with a filtration $\{\mathcal{F}_t\}$ is called a **Standard Brownian Motion** (or Wiener process) if it satisfies the following four properties.

1. **Initial value.** $W_0 = 0$ almost surely.
2. **Independent increments.** For any $0 \le s < t$, the increment $W_t - W_s$ is independent of $\mathcal{F}_s$, the information generated by observing W up to time s. Equivalently, the future movements of the process are independent of its past history.
3. **Stationary Gaussian increments.** For any $0 \le s < t$,

   $$W_t - W_s \sim \mathcal{N}(0, t - s).$$

   The distribution of the increment depends only on the length $t - s$ of the interval, not on its starting point s. This is what *stationary* means here.

4. **Continuous paths.** The sample path $t \mapsto W_t(\omega)$ is a continuous function of t for almost every $\omega \in \Omega$.

These four properties together completely characterize Standard Brownian Motion.

3.2. Unpacking Each Property

Property 1: Starting at Zero

This is a normalization. We fix the process to start at the origin. In finance this corresponds to measuring cumulative returns starting from zero at time 0.

Property 2: Independent Increments

This says that knowing the entire history of W up to time s gives no information about its future movements. The increment $W_t - W_s$ is completely unpredictable given the past.

This is the mathematical expression of the **efficient market hypothesis**: past price movements contain no information about future movements.

Note carefully what independence means here. If we observe $W_1 = 0.5$, we still have no idea whether $W_2 - W_1$ will be positive or negative. The starting level W_s does not influence the distribution of $W_t - W_s$.

Property 3: Stationary Gaussian Increments

This says two things simultaneously.

First, the increment is *Gaussian*. This is the Gaussian structure inherited from the CLT: by the scaling argument, many small random steps sum to a Gaussian variable.

Second, the distribution is *stationary*: it depends only on $t-s$, not on the absolute values of s and t. An increment over a one-hour window has the same distribution whether the window is $[0, 1]$ or $[7, 8]$.

The variance $t - s$ grows with the length of the time interval. A longer interval means more uncertainty about the change in W.

Property 4: Continuous Paths

Brownian paths are continuous functions of time. This distinguishes Brownian motion from processes with jumps, such as Poisson processes, which have discontinuous paths.

Continuity means that the process cannot teleport: it must pass through all intermediate values when moving from one level to another.

3.3. Increments and Their Distribution

For any $0 \le s < t$:

- $E[W_t - W_s] = 0$: increments have zero mean, so Brownian motion has no drift.
- $\mathrm{Var}(W_t - W_s) = t - s$: variance grows linearly with time.

- $SD(W_t - W_s) = \sqrt{t-s}$: standard deviation grows as $\sqrt{t-s}$.

This is the continuous-time analogue of the $\sqrt{n}$ scaling seen for random walks.

3.4. Marginal Distribution

Taking $s = 0$:

$$W_t \sim \mathcal{N}(0, t) \qquad \text{for every } t > 0.$$

At time t, the position of the Brownian motion is a zero-mean Gaussian with variance t. The longer we wait, the more spread out the distribution of W_t becomes.

3.5. Joint Distribution and Covariance Structure

For any times $0 < t_1 < t_2 < \cdots < t_k$, the joint distribution of $(W_{t_1}, W_{t_2}, \ldots, W_{t_k})$ is a k-dimensional multivariate Gaussian.

This means the entire finite-dimensional distribution of Brownian motion is Gaussian.

The covariance between W_s and W_t is given by

$$\mathrm{Cov}(W_s, W_t) = \min(s, t).$$

Derivation. Assume $s \le t$. Write $W_t = W_s + (W_t - W_s)$. Then:

$$\mathrm{Cov}(W_s, W_t) = \mathrm{Cov}(W_s, W_s + (W_t - W_s)) = \mathrm{Var}(W_s) + \mathrm{Cov}(W_s, W_t - W_s).$$

By independent increments, W_s and $W_t - W_s$ are independent, so $\mathrm{Cov}(W_s, W_t - W_s) = 0$. Therefore:

$$\mathrm{Cov}(W_s, W_t) = \mathrm{Var}(W_s) = s = \min(s, t).$$

This covariance formula is the basis for pricing path-dependent derivatives: it quantifies how correlated the process is at two different times.

3.6. Key Properties of Brownian Motion

Symmetry

If W_t is a Standard Brownian Motion, then so is $-W_t$. This follows directly because $-W_t$ satisfies all four defining properties. Financially, this means the model is symmetric between upward and downward movements.

Brownian Scaling

For any constant $c > 0$, the rescaled process

$$\tilde{W}_t = \frac{1}{c} W_{c^2 t}$$

is also a Standard Brownian Motion. This is called the **self-similarity** or **scaling invariance** of Brownian motion.

To verify: $\tilde{W}_0 = 0$, and

$$\tilde{W}_t - \tilde{W}_s = \frac{1}{c}(W_{c^2 t} - W_{c^2 s}) \sim \mathcal{N}\left(0, \frac{c^2(t-s)}{c^2}\right) = \mathcal{N}(0, t-s).$$

Scaling invariance means that Brownian motion looks statistically the same at any time scale. A one-second Brownian path has the same statistical structure as a one-year path, after appropriate rescaling. This fractal-like self-similarity is one of the reasons Brownian motion appears throughout nature and finance.

The Martingale Property

A stochastic process $\{M_t\}$ is called a **martingale** with respect to $\{\mathcal{F}_t\}$ if:

1. $E[|M_t|] < \infty$ for all t;
2. $E[M_t \mid \mathcal{F}_s] = M_s$ for all $0 \leq s \leq t$.

The second condition says: given current information, the best prediction of the future value of the process is its current value. A martingale has no predictable trend.

Claim. Standard Brownian Motion is a martingale.

Proof. For $s \leq t$:

$$E[W_t \mid \mathcal{F}_s] = E[W_s + (W_t - W_s) \mid \mathcal{F}_s] = W_s + E[W_t - W_s \mid \mathcal{F}_s].$$

Since $W_t - W_s$ is independent of $\mathcal{F}_s$ and $E[W_t - W_s] = 0$:

$$E[W_t \mid \mathcal{F}_s] = W_s + 0 = W_s. \qquad \square$$

The martingale property is fundamental in finance. Under risk-neutral pricing, discounted asset prices must be martingales. Brownian motion is the natural building block for constructing martingale models of asset prices.

Non-Differentiability of Brownian Paths

One of the most important and surprising properties of Brownian motion is that its sample paths are **continuous everywhere but differentiable nowhere** almost surely.

This means W_t is a continuous function of t, but dW_t/dt does not exist at any point.

Intuition. The derivative would require the ratio $(W_{t+h} - W_t)/h$ to converge as $h \to 0$. But

$$\frac{W_{t+h} - W_t}{h} \sim \mathcal{N}\left(0, \frac{1}{h}\right).$$

As $h \to 0$, the variance $1/h \to \infty$. The ratio becomes increasingly wild rather than converging to a finite limit.

This is the reason we cannot write $dW_t = (\text{something})\, dt$. The "velocity" of Brownian motion is infinite almost everywhere. This is why ordinary calculus, which relies on derivatives existing, cannot handle Brownian paths.

Non-differentiability forces us to develop an entirely new theory of integration for stochastic processes. This theory is Itô calculus.

4. Total Variation and Quadratic Variation

Quadratic variation is the concept that most sharply distinguishes Brownian motion from smooth deterministic functions. We introduce it carefully here, contrasting it with the more familiar notion of total variation.

4.1. Total Variation

For a function $f : [0, T] \to \mathbb{R}$, the **total variation** over a partition $\Pi = \{0 = t_0 < t_1 < \cdots < t_n = T\}$ is

$$TV(f, \Pi) = \sum_{k=0}^{n-1} |f(t_{k+1}) - f(t_k)|.$$

The total variation of f on $[0, T]$ is

$$TV(f) = \sup_{\Pi} TV(f, \Pi),$$

where the supremum is over all partitions.

For a smooth function (e.g., $f(t) = \sin t$), the total variation is finite: it measures the total distance travelled by the function.

Brownian motion has infinite total variation almost surely.

Intuition. Brownian motion oscillates extremely rapidly. The cumulative distance it travels grows without bound as we look at finer and finer time scales. Formally, one can show that $TV(W) = +\infty$ a.s. on any interval $[0, T]$.

This is another reason ordinary integration fails. The Riemann–Stieltjes integral $\int f\, dg$ requires g to have finite total variation. Since Brownian motion has infinite total variation, we cannot integrate in the classical Riemann–Stieltjes sense.

4.2. Quadratic Variation

Rather than summing absolute increments, the **quadratic variation** sums squared increments:

$$QV(f, \Pi) = \sum_{k=0}^{n-1} \big(f(t_{k+1}) - f(t_k)\big)^2.$$

The quadratic variation of f on $[0, T]$ is the limit as the mesh $\|\Pi\| = \max_k (t_{k+1} - t_k) \to 0$:

$$[f]_T = \lim_{\|\Pi\| \to 0} \sum_{k=0}^{n-1} \big(f(t_{k+1}) - f(t_k)\big)^2.$$

4.3. Quadratic Variation of Smooth Functions

For a smooth function f, the increment satisfies

$$f(t_{k+1}) - f(t_k) \approx f'(t_k)\, \Delta t_k,$$

so the squared increment is of order $(\Delta t_k)^2$. Summing over all intervals:

$$\sum_{k=0}^{n-1} \big(f(t_{k+1}) - f(t_k)\big)^2 \approx \sum_{k=0}^{n-1} (f'(t_k))^2 (\Delta t_k)^2 \le \|\Pi\| \cdot \sum_{k=0}^{n-1} (f'(t_k))^2 \Delta t_k \to 0$$

as $\|\Pi\| \to 0$, since the second factor converges to $\int_0^T (f'(t))^2\, dt < \infty$.

Therefore, smooth functions have **zero quadratic variation**: $[f]_T = 0$.

4.4. Quadratic Variation of Brownian Motion

For Brownian motion, the situation is entirely different.

Theorem (Lévy). For Standard Brownian Motion,

$$[W]_T = T \quad \text{almost surely.}$$

More precisely, for any sequence of partitions Π_n with $\|\Pi_n\| \to 0$,

$$\sum_{k=0}^{n-1} (W_{t_{k+1}} - W_{t_k})^2 \xrightarrow{L^2} T.$$

This is a deep and remarkable result. The quadratic variation of Brownian motion is deterministic (it equals T with probability one) even though the paths are highly random.

4.5. Proof Sketch of $[W]_T = T$

Let $\Pi = \{0 = t_0 < t_1 < \cdots < t_n = T\}$ be any partition. Define

$$Q_n = \sum_{k=0}^{n-1} (W_{t_{k+1}} - W_{t_k})^2.$$

Step 1: Compute $E[Q_n]$.

Since $W_{t_{k+1}} - W_{t_k} \sim \mathcal{N}(0, \Delta t_k)$, we have $E[(W_{t_{k+1}} - W_{t_k})^2] = \Delta t_k$. Therefore:

$$E[Q_n] = \sum_{k=0}^{n-1} \Delta t_k = T.$$

Step 2: Compute $\mathrm{Var}(Q_n)$.

By independence of increments over disjoint intervals, the squared increments are independent. For a $\mathcal{N}(0, \sigma^2)$ variable Z, $\mathrm{Var}(Z^2) = 2\sigma^4$. Therefore:

$$\mathrm{Var}(Q_n) = \sum_{k=0}^{n-1} 2(\Delta t_k)^2 \le 2\|\Pi\| \sum_{k=0}^{n-1} \Delta t_k = 2\|\Pi\| \cdot T \to 0$$

as $\|\Pi\| \to 0$.

Step 3: Conclude.

Since $E[Q_n] = T$ and $\mathrm{Var}(Q_n) \to 0$, we have $Q_n \to T$ in L^2 (and almost surely along suitable subsequences).

4.6. The Informal Identity $dW_t^2 = dt$

The quadratic variation result $[W]_T = T$ is written in differential form as

$$(dW_t)^2 = dt.$$

This is an informal but extremely useful shorthand. It should be understood as a statement about the accumulation of squared increments:

$$\sum_k (dW_{t_k})^2 \approx \sum_k dt_k = T.$$

In contrast, for a smooth function f:

$$(df_t)^2 = (f'(t))^2(dt)^2 \approx 0,$$

since $(dt)^2$ is negligible compared to dt.

The fundamental contrast is:

Process	Quadratic variation	Differential form
Smooth function $f(t)$	$[f]_T = 0$	$(df)^2 = 0$
Brownian motion W_t	$[W]_T = T$	$(dW)^2 = dt$

4.7. The Itô Multiplication Table

Beyond the identity $(dW_t)^2 = dt$, we also have:

- $dW_t \cdot dt = 0$, because $E[(dW_t \cdot dt)^2] = (dt)^3 \to 0$ faster.
- $(dt)^2 = 0$, for the same reason.

These three rules form the **Itô multiplication table**:

$\times$	dW_t	dt
dW_t	dt	0
dt	0	0

These rules are used mechanically in every application of Itô's Lemma. They tell us which terms in a Taylor expansion survive and which can be discarded.

5. The Itô Stochastic Integral

Before stating Itô's Lemma, we need to understand what it means to integrate with respect to Brownian motion. This is the Itô stochastic integral, and it is a concept that does not appear in ordinary calculus.

5.1. Why We Need a New Integral

In ordinary calculus, if $g : [0,T] \to \mathbb{R}$ is a function, the Riemann–Stieltjes integral $\int_0^T f(t)\,dg(t)$ is well defined when g has finite total variation.

As we established, Brownian motion has infinite total variation. Therefore, the ordinary Riemann–Stieltjes construction breaks down, and we need a different definition of $\int_0^T f(t)\,dW_t$.

5.2. Intuitive Construction

The Itô integral is constructed as a limit of Riemann-type sums, but with a crucial choice: we always evaluate the integrand at the **left endpoint** of each interval.

Given a partition $0 = t_0 < t_1 < \cdots < t_n = T$, define the approximating sum:

$$I_n = \sum_{k=0}^{n-1} f(t_k) \cdot (W_{t_{k+1}} - W_{t_k}).$$

The key feature is that $f(t_k)$ is evaluated at t_k, which is $\mathcal{F}_{t_k}$-measurable (it is known before the increment $W_{t_{k+1}} - W_{t_k}$ is revealed). This is called the **non-anticipating** or **adapted** property.

As the mesh $\|\Pi\| \to 0$, the sum I_n converges in L^2 to the Itô integral:

$$\int_0^T f(t)\, dW_t = \lim_{n\to\infty} I_n.$$

5.3. Why Left Endpoints?

The choice of left endpoints is not arbitrary. If we used, say, mid-points $\frac{1}{2}(t_k + t_{k+1})$, we would get a different integral (the Stratonovich integral), which obeys the ordinary chain rule instead of Itô's Lemma.

The left-endpoint choice makes the integral non-anticipating: at the time we commit to a position (choose $f(t_k)$), the future Brownian increment is still unknown. This is precisely the structure of a trading strategy: you decide how many shares to hold at time t_k before seeing the price movement from t_k to t_{k+1}.

5.4. Key Properties of the Itô Integral

Let $I_t = \int_0^t f(s)\, dW_s$ for an adapted process f satisfying $E[\int_0^T f(s)^2\, ds] < \infty$. Then:

- **Zero mean.** $E[I_t] = 0$. The Itô integral is always a zero-mean process.
- **Itô Isometry.**
$$E\left[\left(\int_0^T f(s)\, dW_s\right)^2\right] = E\left[\int_0^T f(s)^2\, ds\right].$$
  This fundamental identity relates the L^2 norm of the stochastic integral to the ordinary integral of the squared integrand. It is used constantly in proofs.
- **Martingale property.** I_t is a martingale. $E[I_t \mid \mathcal{F}_s] = I_s$ for $s \le t$.
- **Linearity.** $\int_0^T (\alpha f + \beta g)\, dW = \alpha \int_0^T f\, dW + \beta \int_0^T g\, dW$.

The martingale property of the Itô integral is one of its most important features. Any process of the form $\int_0^t f(s)\, dW_s$ is a (local) martingale. This is why the Itô integral is the natural tool for constructing martingale models in finance.

6. Itô's Lemma

Itô's Lemma is the chain rule of stochastic calculus. It is the single most important computational tool in the subject.

6.1. Motivation: Why Ordinary Calculus Fails

In ordinary calculus, the chain rule states: if $Y = f(X)$ and X is a smooth function of t, then

$$dY = f'(X)\,dX.$$

Equivalently, by Taylor's theorem to first order:

$$f(X + dX) - f(X) = f'(X)\,dX + \frac{1}{2}f''(X)\,(dX)^2 + \cdots$$

For a smooth deterministic path, $(dX)^2 \sim (dt)^2 \to 0$ faster than dt, so the second-order term is negligible and the chain rule $dY = f'(X)\,dX$ holds.

However, when $X = W_t$ is Brownian motion, $(dW_t)^2 = dt \neq 0$. The second-order term is of the same order as the first-order term and must be retained. Discarding it would give the wrong answer.

6.2. Itô's Lemma: One-Dimensional Version

Theorem (Itô's Lemma). Let W_t be Standard Brownian Motion and let $f : \mathbb{R} \to \mathbb{R}$ be a twice continuously differentiable function. Then:

$$df(W_t) = f'(W_t)\,dW_t + \frac{1}{2}f''(W_t)\,dt.$$

In integral form:

$$f(W_t) = f(W_0) + \int_0^t f'(W_s)\,dW_s + \frac{1}{2}\int_0^t f''(W_s)\,ds.$$

The term $\frac{1}{2}f''(W_t)\,dt$ is the **Itô correction term**. It has no analogue in ordinary calculus and arises entirely because $(dW_t)^2 = dt$.

6.3. Intuitive Derivation

Apply the second-order Taylor expansion:

$$f(W_{t+dt}) - f(W_t) \approx f'(W_t)\,(W_{t+dt} - W_t) + \frac{1}{2}f''(W_t)\,(W_{t+dt} - W_t)^2 + O(|dW_t|^3).$$

Write $dW_t = W_{t+dt} - W_t$ and apply the Itô rule $(dW_t)^2 = dt$:

$$df(W_t) = f'(W_t)\,dW_t + \frac{1}{2}f''(W_t)\,dt.$$

Higher-order terms vanish: $|dW_t|^3 \sim |dt|^{3/2} \to 0$ faster than dt.

6.4. Itô's Lemma for General SDEs

More generally, let X_t satisfy the SDE

$$dX_t = \mu(t, X_t)\, dt + \sigma(t, X_t)\, dW_t,$$

and let $f(t, x)$ be twice continuously differentiable in x and once in t. Then:

$$df(t, X_t) = \frac{\partial f}{\partial t}\, dt + \frac{\partial f}{\partial x}\, dX_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}\, (dX_t)^2.$$

Expanding $(dX_t)^2$ using the Itô multiplication table:

$$(dX_t)^2 = (\mu\, dt + \sigma\, dW_t)^2 = \mu^2 (dt)^2 + 2\mu\sigma\, dt\, dW_t + \sigma^2 (dW_t)^2 = \sigma^2\, dt.$$

Substituting back:

$$\boxed{df(t, X_t) = \left(\frac{\partial f}{\partial t} + \mu\frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2\frac{\partial^2 f}{\partial x^2}\right) dt + \sigma\frac{\partial f}{\partial x}\, dW_t.}$$

This is the full general form of Itô's Lemma. The operator

$$\mathcal{L}f = \frac{\partial f}{\partial t} + \mu\frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2\frac{\partial^2 f}{\partial x^2}$$

is called the **generator** of the SDE and appears throughout stochastic control and PDE theory.

6.5. Worked Examples

Example 1: $f(W_t) = W_t^2$

Let $f(x) = x^2$, so $f'(x) = 2x$ and $f''(x) = 2$. Itô's Lemma gives:

$$d(W_t^2) = 2W_t\, dW_t + \frac{1}{2} \cdot 2\, dt = 2W_t\, dW_t + dt.$$

In integral form:

$$W_t^2 = W_0^2 + 2\int_0^t W_s\, dW_s + t = 2\int_0^t W_s\, dW_s + t.$$

Rearranging:

$$\int_0^t W_s\, dW_s = \frac{W_t^2 - t}{2}.$$

In ordinary calculus, $\int_0^t x\, dx = x^2/2$. Here we get $W_t^2/2$ minus the Itô correction $t/2$. The correction arises because $(dW)^2 = dt \neq 0$.

Taking expectations: $E[\int_0^t W_s\, dW_s] = 0$, which is consistent with the Itô integral having zero mean.

Example 2: $f(W_t) = e^{W_t}$

Let $f(x) = e^x$, so $f'(x) = f''(x) = e^x$. Itô's Lemma gives:

$$d(e^{W_t}) = e^{W_t}\,dW_t + \frac{1}{2}e^{W_t}\,dt.$$

In integral form:

$$e^{W_t} = 1 + \int_0^t e^{W_s}\,dW_s + \frac{1}{2}\int_0^t e^{W_s}\,ds.$$

Example 3: The Log of GBM

Let $S_t = S_0 e^{\sigma W_t + \mu t}$ and let $f(t,x) = \ln x$, so $\partial f/\partial x = 1/x$ and $\partial^2 f/\partial x^2 = -1/x^2$.

If S_t satisfies $dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$, then applying Itô's Lemma to $f(t, S_t) = \ln S_t$:

$$d(\ln S_t) = \frac{1}{S_t}\,dS_t - \frac{1}{2}\cdot\frac{1}{S_t^2}\,\sigma^2 S_t^2\,dt = \mu\,dt + \sigma\,dW_t - \frac{\sigma^2}{2}\,dt = \left(\mu - \frac{\sigma^2}{2}\right)dt + \sigma\,dW_t.$$

Integrating:

$$\ln S_t = \ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t,$$

which gives the explicit solution:

$$S_t = S_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t\right).$$

The term $-\sigma^2/2$ is the Itô correction. Without Itô's Lemma, one might incorrectly write $S_t = S_0 e^{\mu t + \sigma W_t}$, which gives the wrong expected value:

$$E[S_t] = S_0 e^{\mu t + \sigma^2 t/2} \neq S_0 e^{\mu t}.$$

The correct formula gives $E[S_t] = S_0 e^{\mu t}$ only when the $-\sigma^2/2$ correction is included in the exponent.

7. Stochastic Differential Equations

A **stochastic differential equation** (SDE) is an equation of the form

$$dX_t = \mu(t, X_t)\,dt + \sigma(t, X_t)\,dW_t, \qquad X_0 = x_0.$$

This is the fundamental model of a continuous-time stochastic process. Understanding SDEs requires understanding each component carefully.

7.1. Components of an SDE

The Drift Term: $\mu(t, X_t)\,dt$

The drift coefficient $\mu(t, X_t)$ determines the average direction of movement over a small time interval $[t, t + dt]$:

$$E[X_{t+dt} - X_t \mid \mathcal{F}_t] \approx \mu(t, X_t)\,dt.$$

If $\mu > 0$, the process tends to increase. If $\mu < 0$, it tends to decrease. The drift is the deterministic component of the dynamics.

The Diffusion Term: $\sigma(t, X_t)\, dW_t$

The diffusion coefficient $\sigma(t, X_t)$ controls the magnitude of random fluctuations:

$$\mathrm{Var}(X_{t+dt} - X_t \mid \mathcal{F}_t) \approx \sigma(t, X_t)^2\, dt.$$

The term $dW_t \sim \mathcal{N}(0, dt)$ injects Gaussian randomness at each instant. A larger $|\sigma|$ means more volatility. When $\sigma = 0$, the SDE reduces to an ordinary differential equation.

The Discretized Form

Over a small time step $[t, t + \Delta t]$:

$$X_{t+\Delta t} \approx X_t + \mu(t, X_t)\, \Delta t + \sigma(t, X_t)\, \Delta W_t,$$

where $\Delta W_t = W_{t+\Delta t} - W_t \sim \mathcal{N}(0, \Delta t)$.

This is exactly the Euler–Maruyama scheme.

7.2. Existence and Uniqueness

Under Lipschitz and linear growth conditions on μ and σ (which are satisfied by all models we consider), the SDE has a unique strong solution. A **strong solution** is a process X_t that is adapted to the filtration of W_t and satisfies the integral equation

$$X_t = x_0 + \int_0^t \mu(s, X_s)\, ds + \int_0^t \sigma(s, X_s)\, dW_s$$

for all $t \geq 0$ almost surely.

7.3. Example: Geometric Brownian Motion

The SDE for Geometric Brownian Motion (GBM) is

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t.$$

Here both the drift and diffusion are proportional to the current level S_t. This means the percentage change dS_t/S_t is the same regardless of the level of S_t: returns, not absolute price changes, are modelled as Gaussian. This is the standard model for stock prices in the Black–Scholes framework.

The explicit solution, derived via Itô's Lemma above, is:

$$S_t = S_0 \exp\left(\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t \right).$$

Note that $S_t > 0$ for all t if $S_0 > 0$, so GBM never goes negative — an important feature for a stock price model.

7.4. Example: The Ornstein–Uhlenbeck Process

The Ornstein–Uhlenbeck (OU) SDE is

$$dX_t = -\theta X_t\,dt + \sigma\,dW_t, \qquad \theta > 0.$$

The drift term $-\theta X_t\,dt$ is mean-reverting: it pulls the process back toward zero whenever $X_t \neq 0$. If $X_t > 0$, the drift is negative (pointing toward zero). If $X_t < 0$, the drift is positive (again pointing toward zero).

The parameter $\theta > 0$ is the **speed of mean reversion**: a larger θ means faster return to zero.

The explicit solution is:

$$X_t = X_0 e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)}\,dW_s.$$

This can be verified by applying Itô's Lemma to $e^{\theta t} X_t$.

The OU process is stationary in the long run. As $t \to \infty$:

$$X_t \to \mathcal{N}\left(0, \frac{\sigma^2}{2\theta}\right) \quad \text{in distribution.}$$

The OU process is used to model:

- interest rates (Vasicek model);
- volatility in stochastic volatility models;
- commodity prices that revert to a long-run mean;
- the velocity of a Brownian particle subject to friction.

8. The Euler–Maruyama Method

Most SDEs do not have closed-form solutions. The Euler–Maruyama method is the simplest and most widely used numerical scheme for simulating SDE paths.

8.1. Scheme Definition

Fix a time horizon T and divide it into N equal steps of size $\Delta t = T/N$. Define the time grid $t_k = k\,\Delta t$ for $k = 0, 1, \ldots, N$.

Starting from $X_0 = x_0$, iterate:

$$X_{t_{k+1}} = X_{t_k} + \mu(t_k, X_{t_k})\,\Delta t + \sigma(t_k, X_{t_k})\,\Delta W_k,$$

where $\Delta W_k = W_{t_{k+1}} - W_{t_k} \sim \mathcal{N}(0, \Delta t)$ are independent Gaussian increments.

In code, ΔW_k is generated as `np.random.normal(0, np.sqrt(dt))`.

8.2. Why It Works

The scheme approximates the true SDE by treating μ and σ as constant over each small interval. This is valid when Δt is small, because both μ and σ are slowly varying compared to the timescale Δt.

Over the interval $[t_k, t_{k+1}]$, the true SDE says:

$$X_{t_{k+1}} - X_{t_k} = \int_{t_k}^{t_{k+1}} \mu(s, X_s)\, ds + \int_{t_k}^{t_{k+1}} \sigma(s, X_s)\, dW_s \approx \mu(t_k, X_{t_k})\, \Delta t + \sigma(t_k, X_{t_k})\, \Delta W_k,$$

where the approximation freezes the coefficients at their values at the left endpoint.

8.3. Convergence Orders

The Euler–Maruyama scheme has two notions of convergence:

- **Strong convergence (pathwise).** The scheme approximates individual sample paths:

$$E\left[\max_k |X_{t_k} - \hat{X}_{t_k}|\right] = O(\sqrt{\Delta t}).$$

  The strong order is 1/2: halving Δt reduces pathwise error by approximately $1/\sqrt{2}$.

- **Weak convergence (distributional).** The scheme approximates the distribution of X_T:

$$|E[g(X_T)] - E[g(\hat{X}_T)]| = O(\Delta t)$$

  for smooth test functions g. The weak order is 1: halving Δt halves the distributional error.

For option pricing (computing $E[g(X_T)]$), weak convergence suffices and the scheme is first-order accurate. For path-dependent quantities (barrier options, Asian options), strong convergence matters and the scheme is slower (order 1/2).

Higher-order schemes (Milstein, Runge–Kutta) achieve better convergence rates but are more complex to implement.

8.4. Stability

For the OU process $dX_t = -\theta X_t\, dt + \sigma\, dW_t$, the Euler–Maruyama scheme requires $\Delta t < 2/\theta$ for stability. If Δt is too large relative to the mean-reversion speed, the discrete scheme can become unstable and produce exploding paths.

This is a general principle: stiff SDEs (with large θ) require small time steps.

9. Simulation in Python

This section provides complete, well-commented Python code for each key computation of this week. Each snippet is self-contained and can be run directly.

9.1. Libraries Used

```python
import numpy as np                # numerical arrays and random number
    generation
import matplotlib.pyplot as plt # plotting
from scipy.stats import norm     # Gaussian density, CDF
```

9.2. Simulating a Single Brownian Path

The key idea: Brownian increments are iid Gaussian, so we generate them directly and take a cumulative sum to build the path.

```python
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(0)

T = 1.0       # time horizon
N = 1000      # number of time steps
dt = T / N    # step size

# Generate N independent Brownian increments dW_k ~ N(0, dt)
dW = np.random.normal(0, np.sqrt(dt), size=N)

# Build the path: W_0 = 0, then W_{k+1} = W_k + dW_k
# np.cumsum gives [dW_0, dW_0+dW_1, ...] -- prefix it with 0
W = np.concatenate([[0], np.cumsum(dW)])

t = np.linspace(0, T, N + 1)

plt.figure(figsize=(10, 4))
plt.plot(t, W, linewidth=0.8, color='steelblue')
plt.xlabel('$t$')
plt.ylabel('$W_t$')
plt.title('Single Standard Brownian Motion Path')
plt.axhline(0, color='black', linewidth=0.5, linestyle='--')
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('outputs/q_bm_single.png', dpi=150)
plt.show()
```

9.3. Simulating Multiple Brownian Paths

Simulating many paths simultaneously reveals the spread of the distribution of W_t:

```python
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(0)

T = 1.0
N = 1000
n_paths = 50   # number of independent paths
```

```python
dt = T / N

# Generate all increments at once: shape (n_paths, N)
dW = np.random.normal(0, np.sqrt(dt), size=(n_paths, N))

# Cumulative sum along time axis, prepend zeros for W_0 = 0
W = np.hstack([np.zeros((n_paths, 1)), np.cumsum(dW, axis=1)])

t = np.linspace(0, T, N + 1)

plt.figure(figsize=(10, 5))
for i in range(n_paths):
    plt.plot(t, W[i], alpha=0.3, linewidth=0.6, color='steelblue')

# Overlay the theoretical 2*std band
std = np.sqrt(t)
plt.fill_between(t, -2*std, 2*std, alpha=0.15, color='red',
                 label=r'$\pm 2\sqrt{t}$ band')
plt.xlabel('$t$')
plt.ylabel('$W_t$')
plt.title(f'{n_paths} Brownian Motion Paths with $\\pm 2\\sqrt{{t}}$
    Band')
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('outputs/q_bm_multi.png', dpi=150)
plt.show()
```

The $\pm 2\sqrt{t}$ band corresponds to the ± 2 standard deviation range of $W_t \sim \mathcal{N}(0, t)$. Approximately 95% of paths should remain within this band at each fixed time.

9.4. Verifying the Marginal Distribution

We can verify that $W_t \sim \mathcal{N}(0, t)$ empirically by plotting the histogram of simulated values at a fixed time t:

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

np.random.seed(0)

T = 1.0
N = 1000
n_paths = 10000   # large number for accurate histogram
dt = T / N

dW = np.random.normal(0, np.sqrt(dt), size=(n_paths, N))
W = np.hstack([np.zeros((n_paths, 1)), np.cumsum(dW, axis=1)])

# Look at W at a fixed time, say t = 0.5
t_fixed = 0.5
idx = int(t_fixed / dt)  # index corresponding to t_fixed
W_at_t = W[:, idx]

x = np.linspace(-4 * np.sqrt(t_fixed), 4 * np.sqrt(t_fixed), 300)
```

```

plt.figure(figsize=(7, 4))
plt.hist(W_at_t, bins=60, density=True, alpha=0.6, color='steelblue'
    ,
         label=f'Simulated $W_{{t}}$ at $t={t_fixed}$')
plt.plot(x, norm.pdf(x, 0, np.sqrt(t_fixed)), 'r-', lw=2,
         label=r'$\mathcal{N}(0, t)$ density')
plt.xlabel('$W_t$')
plt.ylabel('Density')
plt.title(f'Marginal Distribution of $W_t$ at $t = {t_fixed}$')
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('outputs/q_bm_marginal.png', dpi=150)
plt.show()

print(f'Empirical mean:     {np.mean(W_at_t):.4f}  (theory: 0)')
print(f'Empirical variance: {np.var(W_at_t):.4f}  (theory: {t_fixed
    })')
```

9.5. Empirically Verifying Quadratic Variation

The following code computes the quadratic variation of a simulated Brownian path and compares it with the theoretical value T:

```
import numpy as np

np.random.seed(0)

T = 1.0
N = 10000   # fine partition for accurate QV estimate
dt = T / N

dW = np.random.normal(0, np.sqrt(dt), size=N)
W = np.concatenate([[0], np.cumsum(dW)])

# Quadratic variation: sum of squared increments
increments = np.diff(W)          # dW_k = W_{k+1} - W_k
QV = np.sum(increments**2)       # sum of (dW_k)^2

print(f'Empirical quadratic variation : {QV:.6f}')
print(f'Theoretical value T           : {T:.6f}')
print(f'Difference                    : {abs(QV - T):.6f}')

# Compare with total variation (much larger)
TV = np.sum(np.abs(increments))
print(f'\nEmpirical total variation     : {TV:.4f}  (expected:
    infinite)')
```

The empirical QV should be close to $T = 1$, confirming $[W]_T = T$. The total variation will be very large, confirming that Brownian motion has infinite total variation.

9.6. QV Convergence as Partition Becomes Finer

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(0)

T = 1.0
n_steps_list = [5, 10, 25, 50, 100, 500, 1000, 5000, 10000]
qv_list = []

for N in n_steps_list:
    dt = T / N
    dW = np.random.normal(0, np.sqrt(dt), size=N)
    QV = np.sum(dW**2)
    qv_list.append(QV)

plt.figure(figsize=(8, 4))
plt.semilogx(n_steps_list, qv_list, 'bo-', markersize=5,
             label='Empirical QV')
plt.axhline(T, color='r', linestyle='--', linewidth=1.5,
            label=f'True QV = $T = {T}$')
plt.xlabel('Number of steps $N$ (log scale)')
plt.ylabel('Quadratic variation')
plt.title('Convergence of Quadratic Variation to $T$ as Partition
    Refines')
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('outputs/q_bm_qv_conv.png', dpi=150)
plt.show()
```

As N increases (partition becomes finer), the empirical QV converges to T. Note that for coarser partitions the estimate is noisier — this reflects the $O(\sqrt{\Delta t})$ standard deviation of the estimator.

9.7. Verifying the Itô Result for W_t^2

By Itô's Lemma, $W_t^2 - t$ is a martingale, so $E[W_t^2] = t$ for all t. We verify this empirically:

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(0)

T = 1.0
N = 500
n_paths = 10000
dt = T / N

dW = np.random.normal(0, np.sqrt(dt), size=(n_paths, N))
W = np.hstack([np.zeros((n_paths, 1)), np.cumsum(dW, axis=1)])

t = np.linspace(0, T, N + 1)

# Compute E[W_t^2] across paths at each time step
EW2 = np.mean(W**2, axis=0)
```

```python

plt.figure(figsize=(8, 4))
plt.plot(t, EW2, label=r'Empirical $E[W_t^2]$', color='steelblue')
plt.plot(t, t, 'r--', label=r'Theoretical $E[W_t^2] = t$', linewidth=1.5)
plt.xlabel('$t$')
plt.ylabel(r'$E[W_t^2]$')
plt.title(r"Verification of It\^{o}'s Result: $E[W_t^2] = t$")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('outputs/q_ito_verify.png', dpi=150)
plt.show()

print(f"E[W_T^2] empirical : {np.mean(W[:, -1]**2):.4f}")
print(f"T (theoretical)    : {T:.4f}")
```

9.8. Euler–Maruyama for Geometric Brownian Motion

```python
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(0)

# GBM parameters
S0 = 100.0   # initial stock price
mu = 0.05    # annual drift (5%)
sigma = 0.2  # annual volatility (20%)
T = 1.0      # 1 year
N = 1000     # time steps
n_paths = 20 # paths to plot
dt = T / N

t = np.linspace(0, T, N + 1)

plt.figure(figsize=(10, 5))

for _ in range(n_paths):
    S = np.zeros(N + 1)
    S[0] = S0

    for k in range(N):
        # Euler-Maruyama step: S_{k+1} = S_k + mu*S_k*dt + sigma*S_k*dW_k
        dW = np.random.normal(0, np.sqrt(dt))
        S[k + 1] = S[k] + mu * S[k] * dt + sigma * S[k] * dW

    plt.plot(t, S, alpha=0.5, linewidth=0.8)

# Overlay the theoretical mean E[S_t] = S_0 * exp(mu * t)
plt.plot(t, S0 * np.exp(mu * t), 'r--', linewidth=2,
         label=r'$E[S_t] = S_0 e^{\mu t}$')
plt.xlabel('$t$ (years)')
plt.ylabel('$S_t$')
plt.title('Geometric Brownian Motion via Euler--Maruyama')
```

```python
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('outputs/q_gbm_paths.png', dpi=150)
plt.show()
```

9.9. Euler–Maruyama: Vectorized for Many Paths

Simulating paths one at a time with a Python loop is slow. Vectorizing across paths is much faster:

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import lognorm

np.random.seed(0)

S0 = 100.0
mu = 0.05
sigma = 0.2
T = 1.0
N = 500
n_paths = 50000   # large number for accurate distribution
dt = T / N

# All increments at once: shape (n_paths, N)
dW = np.random.normal(0, np.sqrt(dt), size=(n_paths, N))

S = np.zeros((n_paths, N + 1))
S[:, 0] = S0

# Vectorized Euler-Maruyama: update all paths simultaneously
for k in range(N):
    S[:, k + 1] = (
        S[:, k]
        + mu * S[:, k] * dt
        + sigma * S[:, k] * dW[:, k]
    )

ST = S[:, -1]  # terminal stock prices

# Theoretical distribution of S_T:
# S_T ~ LogNormal with parameters
# log-mean = log(S0) + (mu - sigma^2/2)*T
# log-std  = sigma * sqrt(T)
log_mean = np.log(S0) + (mu - 0.5 * sigma**2) * T
log_std = sigma * np.sqrt(T)

x = np.linspace(ST.min(), ST.max(), 300)

plt.figure(figsize=(8, 4))
plt.hist(ST, bins=100, density=True, alpha=0.6, color='steelblue',
         label='Simulated $S_T$')
plt.plot(x, lognorm.pdf(x, s=log_std, scale=np.exp(log_mean)),
         'r-', lw=2, label='Theoretical log-normal density')
plt.xlabel('$S_T$')
```

```
plt.ylabel('Density')
plt.title('Terminal Stock Price Distribution (GBM, Euler--Maruyama)'
    )
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('outputs/q_gbm_terminal_dist.png', dpi=150)
plt.show()

print(f'Empirical E[S_T]   : {np.mean(ST):.4f}')
print(f'Theoretical E[S_T] : {S0 * np.exp(mu * T):.4f}')
print(f'Empirical Std[S_T] : {np.std(ST):.4f}')
```

9.10. Euler–Maruyama for the Ornstein–Uhlenbeck Process

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

np.random.seed(0)

# OU parameters
X0 = 2.0       # starting value (away from mean)
theta = 2.0    # speed of mean reversion
sigma = 0.5    # diffusion coefficient
T = 5.0        # longer horizon to see reversion
N = 5000
n_paths = 10
dt = T / N

t = np.linspace(0, T, N + 1)

plt.figure(figsize=(10, 5))

for _ in range(n_paths):
    X = np.zeros(N + 1)
    X[0] = X0

    for k in range(N):
        # OU step: X_{k+1} = X_k - theta * X_k * dt + sigma * dW_k
        dW = np.random.normal(0, np.sqrt(dt))
        X[k + 1] = X[k] - theta * X[k] * dt + sigma * dW

    plt.plot(t, X, alpha=0.5, linewidth=0.8)

# Stationary mean (= 0 for this OU) and +/- 2 std band
stationary_std = sigma / np.sqrt(2 * theta)
plt.axhline(0, color='black', linewidth=1.2, linestyle='--',
            label='Stationary mean = 0')
plt.fill_between(t,
                 -2 * stationary_std,
                  2 * stationary_std,
                  alpha=0.1, color='red',
                  label=r'$\pm 2\sigma/\sqrt{2\theta}$ band')
plt.xlabel('$t$')
```

```
plt.ylabel('$X_t$')
plt.title('Ornstein--Uhlenbeck Process (Mean-Reverting)')
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('outputs/q_ou_paths.png', dpi=150)
plt.show()

print(f'Theoretical stationary std: {stationary_std:.4f}')
```

Note how all paths start at $X_0 = 2.0$ but eventually fluctuate around zero with spread approximately $\sigma/\sqrt{2\theta}$, consistent with the stationary distribution $\mathcal{N}(0, \sigma^2/2\theta)$.

10. Connection to Finance

10.1. The Black–Scholes Model

The standard model for stock prices in quantitative finance is Geometric Brownian Motion:

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t.$$

Here:

- μ is the expected rate of return (drift);
- σ is the volatility (diffusion coefficient);
- W_t is a Standard Brownian Motion.

Applying Itô's Lemma to $\ln S_t$ yields the explicit solution:

$$S_t = S_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right) t + \sigma W_t\right).$$

The term $-\sigma^2/2$ is the Itô correction. It ensures that:

$$E[S_t] = S_0 e^{\mu t},$$

which is the correct expected growth. Without the correction, one would overestimate expected stock prices.

The log-return over $[0, T]$ is:

$$\ln\left(\frac{S_T}{S_0}\right) = \left(\mu - \frac{\sigma^2}{2}\right) T + \sigma W_T \sim \mathcal{N}\left(\left(\mu - \frac{\sigma^2}{2}\right) T, \; \sigma^2 T\right).$$

So log-returns are Gaussian under GBM, and their standard deviation grows as $\sigma\sqrt{T}$ — the $\sqrt{T}$ scaling of Brownian motion.

10.2. Why the Itô Correction Matters Financially

Consider two investors who model the same stock with the same μ and σ.

Investor A correctly uses GBM and calculates $E[S_T] = S_0 e^{\mu T}$.

Investor B incorrectly ignores the Itô correction and writes $S_T = S_0 e^{\mu T + \sigma W_T}$. Then:

$$E_B[S_T] = S_0 e^{\mu T} \cdot E[e^{\sigma W_T}] = S_0 e^{\mu T} \cdot e^{\sigma^2 T/2}.$$

Investor B overestimates the expected stock price by a factor of $e^{\sigma^2 T/2}$. For $\sigma = 0.2$ and $T = 10$ years, this factor is $e^{0.2} \approx 1.22$, a 22% overestimate. This is a significant pricing error.

10.3. Realized Variance and Quadratic Variation

For a log-price process $X_t = \ln S_t$ following GBM:

$$dX_t = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma \, dW_t.$$

The quadratic variation of X over $[0, T]$ is:

$$[X]_T = \sigma^2 T.$$

This identity is remarkable: the quadratic variation of the log-price depends only on σ, not on the drift μ. We can therefore estimate σ from high-frequency price data without needing to know or estimate μ.

The **realized variance estimator** is:

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{k=0}^{n-1} \left(\ln S_{t_{k+1}} - \ln S_{t_k}\right)^2.$$

This converges to σ^2 as the partition becomes finer (as $\Delta t \to 0$). This is the theoretical basis for realized volatility measures widely used in risk management and option pricing.

10.4. Martingales and Risk-Neutral Pricing

In financial mathematics, option pricing relies on the concept of a **risk-neutral measure** Q, under which the discounted stock price $e^{-rt} S_t$ is a martingale.

Under Q, the GBM drift changes from μ to r (the risk-free rate):

$$dS_t = r S_t \, dt + \sigma S_t \, dW_t^Q,$$

where W_t^Q is a Brownian motion under Q. This change of measure is performed using the **Girsanov theorem**, which is built on the properties of the Itô integral and martingales.

The fair price of a European option with payoff $g(S_T)$ is then:

$$V_0 = e^{-rT} E^Q[g(S_T)].$$

This is the Black–Scholes pricing formula in abstract form. Everything in it — the SDE, the change of measure, the expectation — is built on Brownian motion and Itô calculus.

11. Key Takeaways

- Brownian motion W_t is the continuous-time limit of the symmetric random walk, obtained by scaling steps as $\pm\sqrt{\Delta t}$ and letting $\Delta t \to 0$.
- A filtration $\{\mathcal{F}_t\}$ encodes the accumulation of information over time. An adapted process depends only on current and past information, never on the future.
- Standard Brownian Motion satisfies four axioms: $W_0 = 0$, independent increments, stationary Gaussian increments $W_t - W_s \sim \mathcal{N}(0, t-s)$, and continuous paths.
- The covariance of Brownian motion is $\mathrm{Cov}(W_s, W_t) = \min(s,t)$.
- Brownian motion is a martingale: $E[W_t \mid \mathcal{F}_s] = W_s$.
- Brownian paths are continuous everywhere but differentiable nowhere, which is why ordinary calculus fails.
- Brownian motion has infinite total variation but finite, non-zero quadratic variation: $[W]_T = T$.
- The Itô multiplication table encodes how products of differentials behave: $(dW)^2 = dt$, $dW \cdot dt = 0$, $(dt)^2 = 0$.
- The Itô stochastic integral $\int f\,dW$ is defined via left-endpoint Riemann sums, has zero mean, and satisfies the Itô isometry.
- Itô's Lemma is the stochastic chain rule: $df(W_t) = f'(W_t)\,dW_t + \frac{1}{2}f''(W_t)\,dt$.
- The extra $\frac{1}{2}f''\,dt$ term arises from $(dW)^2 = dt$ and has no counterpart in ordinary calculus.
- GBM satisfies $dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$ with explicit solution $S_t = S_0 \exp\big((\mu - \sigma^2/2)t + \sigma W_t\big)$.
- The $-\sigma^2/2$ correction in GBM is the Itô correction and is essential for the correct expected value $E[S_t] = S_0 e^{\mu t}$.
- The Euler–Maruyama method discretizes SDEs with strong order 1/2 and weak order 1.
- Quadratic variation of log-prices gives a model-free volatility estimator: the realized variance.

12. Suggested External References

- Bernt Øksendal — *Stochastic Differential Equations: An Introduction with Applications* (accessible and comprehensive)
- Steven Shreve — *Stochastic Calculus for Finance II: Continuous-Time Models* (finance-focused, rigorous)
- Ioannis Karatzas and Steven Shreve — *Brownian Motion and Stochastic Calculus* (mathematically rigorous reference)
- Marek Capiński and Tomasz Zastawniak — *Mathematics for Finance* (more introductory, good for building intuition)
- 3Blue1Brown — *Stochastic Calculus and Brownian Motion* (video series, highly visual and intuitive)

- MIT OpenCourseWare 18.650 — *Statistics for Applications* (background on Gaussian distributions)

Week 3 Summary

This week introduced the mathematical foundation of continuous-time stochastic processes.

We began by constructing Brownian motion as the continuous-time limit of the symmetric random walk, inheriting the Gaussian structure of the CLT but living in continuous time. We introduced filtrations and the notion of an adapted process, which are the language through which information is handled in continuous-time finance.

We gave the four-axiom definition of Standard Brownian Motion and unpacked each property carefully: the independent increment structure, the Gaussian distribution of increments, the covariance formula $\mathrm{Cov}(W_s, W_t) = \min(s, t)$, the martingale property, and the non-differentiability of paths.

We then developed the concept of quadratic variation. While smooth functions have zero quadratic variation, Brownian motion has non-zero quadratic variation $[W]_T = T$, expressed informally as $dW_t^2 = dt$. The contrast with total variation was highlighted: Brownian motion has infinite total variation, ruling out ordinary Riemann–Stieltjes integration, but finite quadratic variation, which is what Itô calculus exploits.

We constructed the Itô stochastic integral intuitively via left-endpoint sums and stated its key properties: zero mean, the Itô isometry, and the martingale property.

Building on these foundations, we derived Itô's Lemma from a Taylor expansion, applying the rule $(dW)^2 = dt$ to retain the second-order term. We applied Itô's Lemma to three key examples: W_t^2, e^{W_t}, and Geometric Brownian Motion.

We introduced Stochastic Differential Equations formally, with careful discussion of drift, diffusion, and the Euler–Maruyama numerical scheme including its convergence orders and stability.

Finally, we connected all these ideas to finance: the Black–Scholes GBM model, the Itô correction and why it matters, realized variance as a model-free volatility estimator, and the martingale structure underlying risk-neutral pricing.

These tools form the core mathematical language of continuous-time quantitative finance. Every major result in the field — option pricing, hedging, risk management, interest rate modelling — is built on the foundations laid this week.